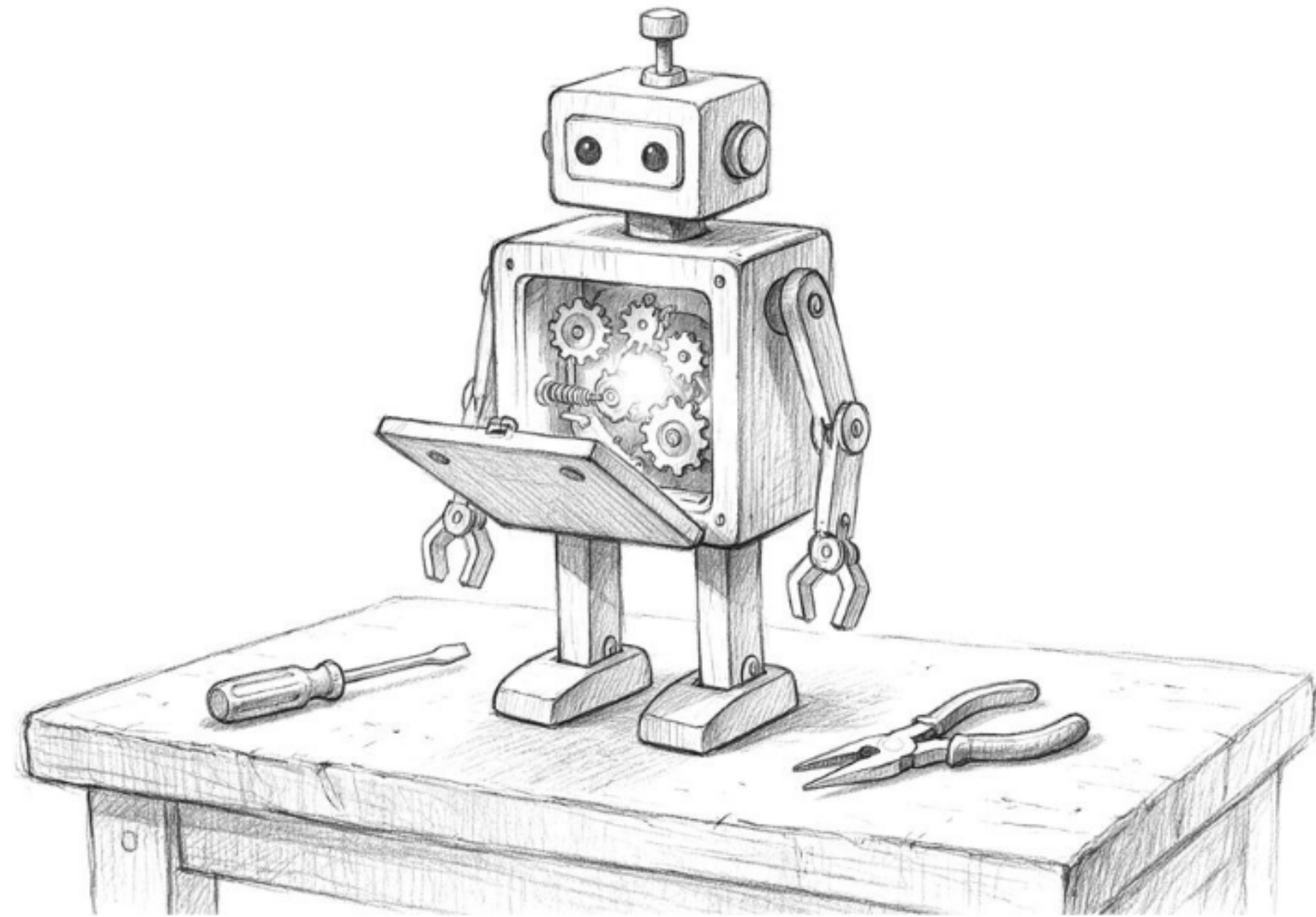


How Machines Actually Learn



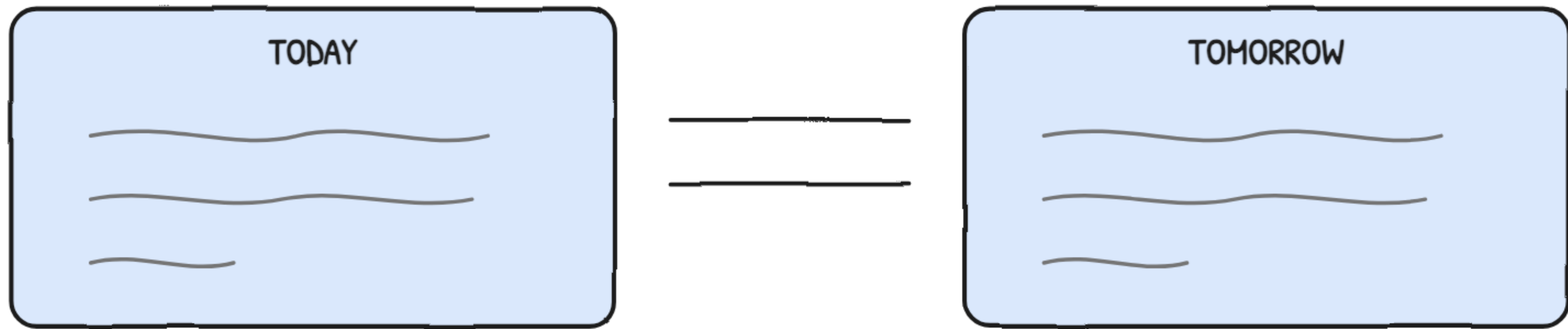
The three ideas inside ChatGPT

Gourav Shah · School of DevOps & AI · Lesson + Lab + Quiz



How did ChatGPT learn to write, and why doesn't it learn from your chats?

Same question tomorrow, same wrong answer



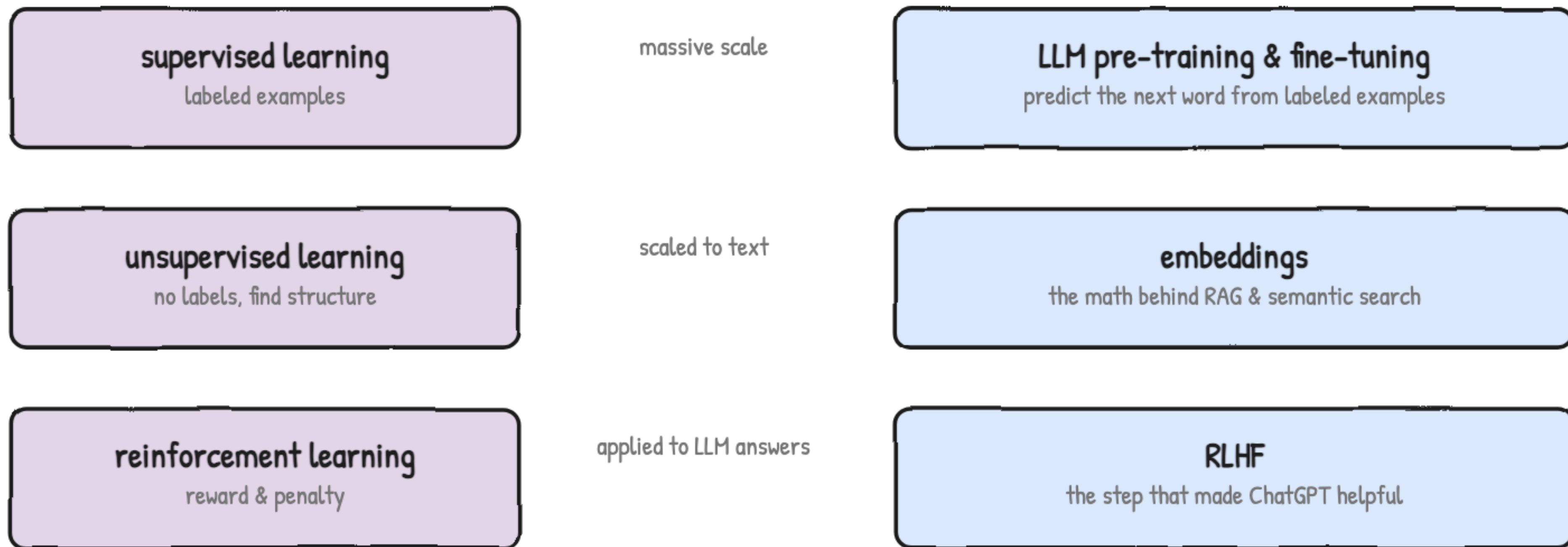
same numbers, every single time — why?

Three ideas explain why



Three ways a machine learns

Every modern AI system scales one of three learning types



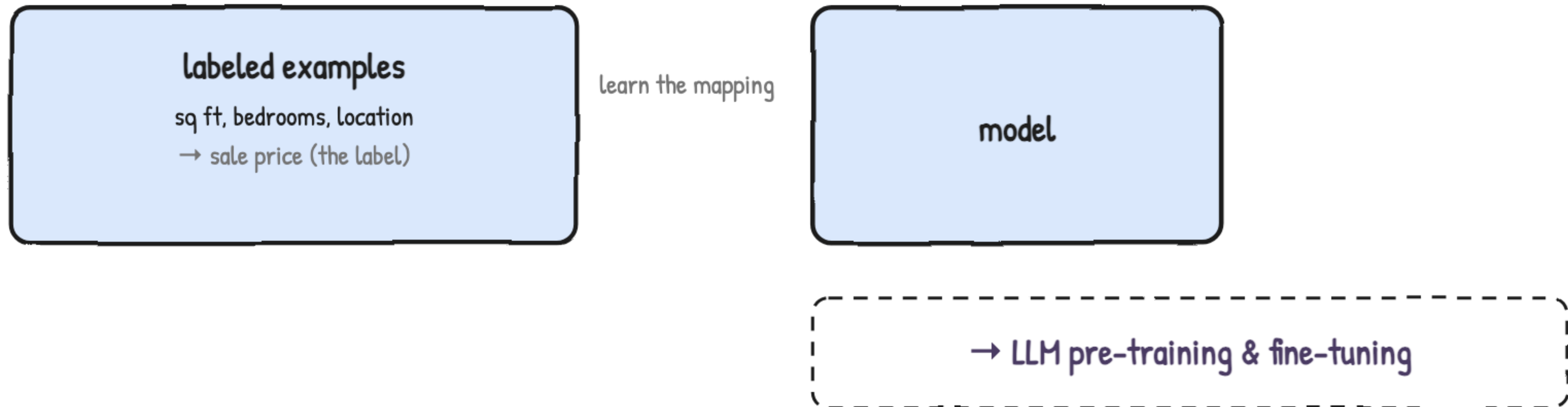
Name the pattern small, and you'll spot it inside ChatGPT

Training a new hire on ten thousand solved tickets



Read enough examples, and the pattern starts to stick

Supervised learning: show the answer, learn the mapping



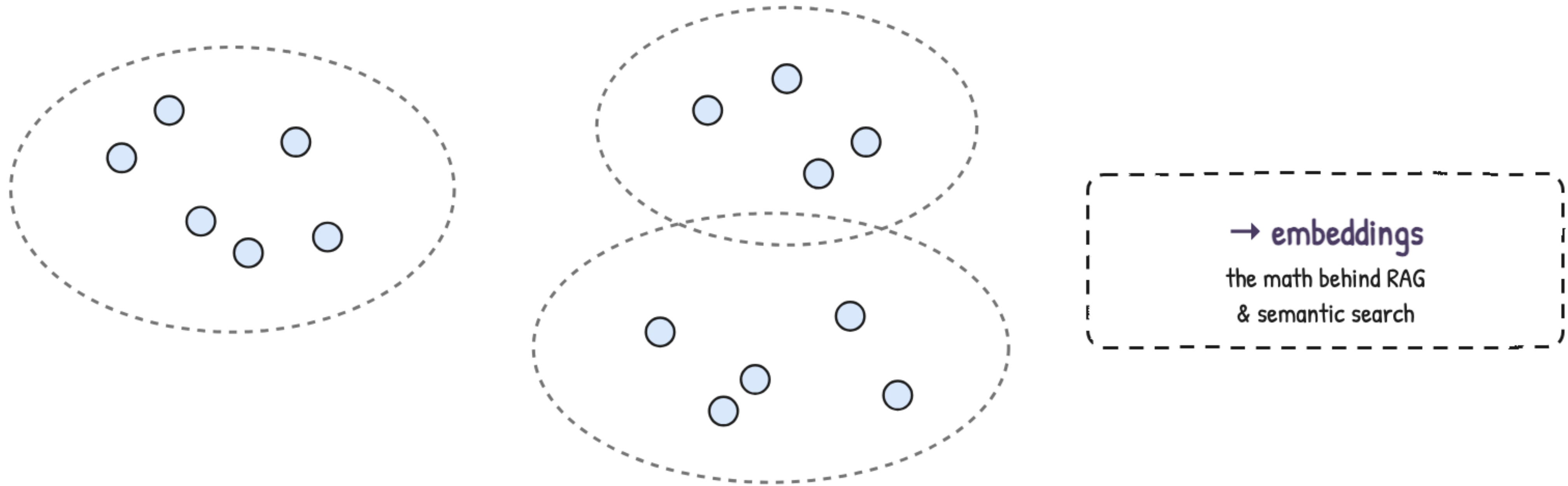
This is how ChatGPT gets pre-trained and fine-tuned

Ten thousand customer records, no categories attached



Left alone, the groups still surface

Unsupervised learning: no labels, just structure



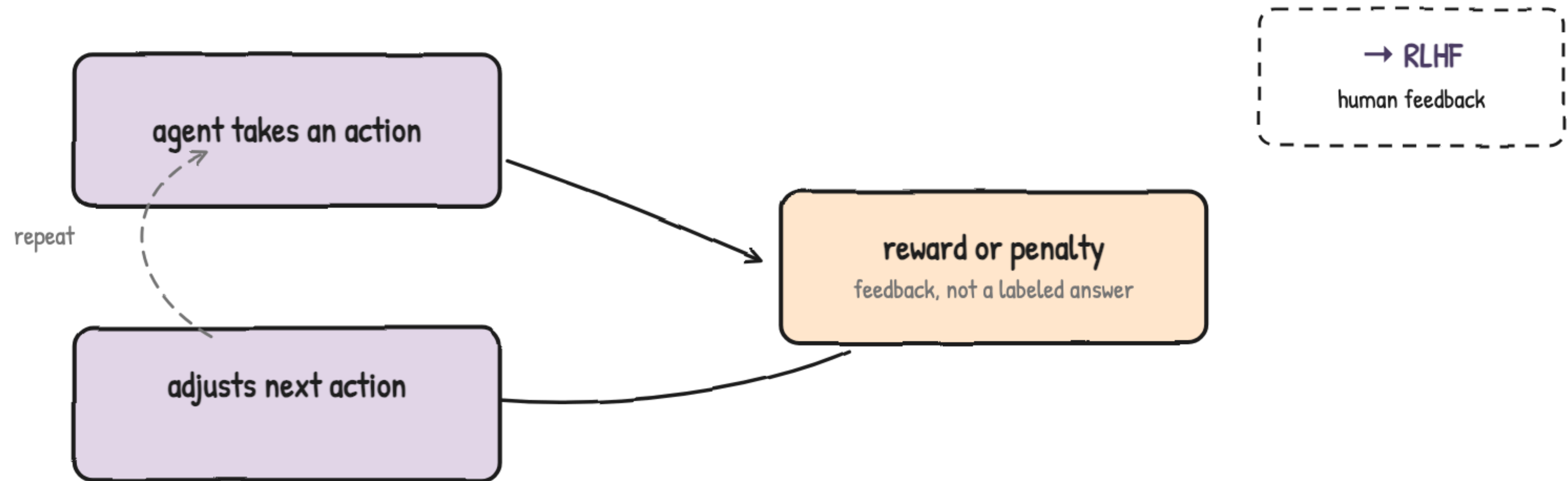
This becomes embeddings — the math behind RAG and search

Treats, corrections, and a dog that learns to sit

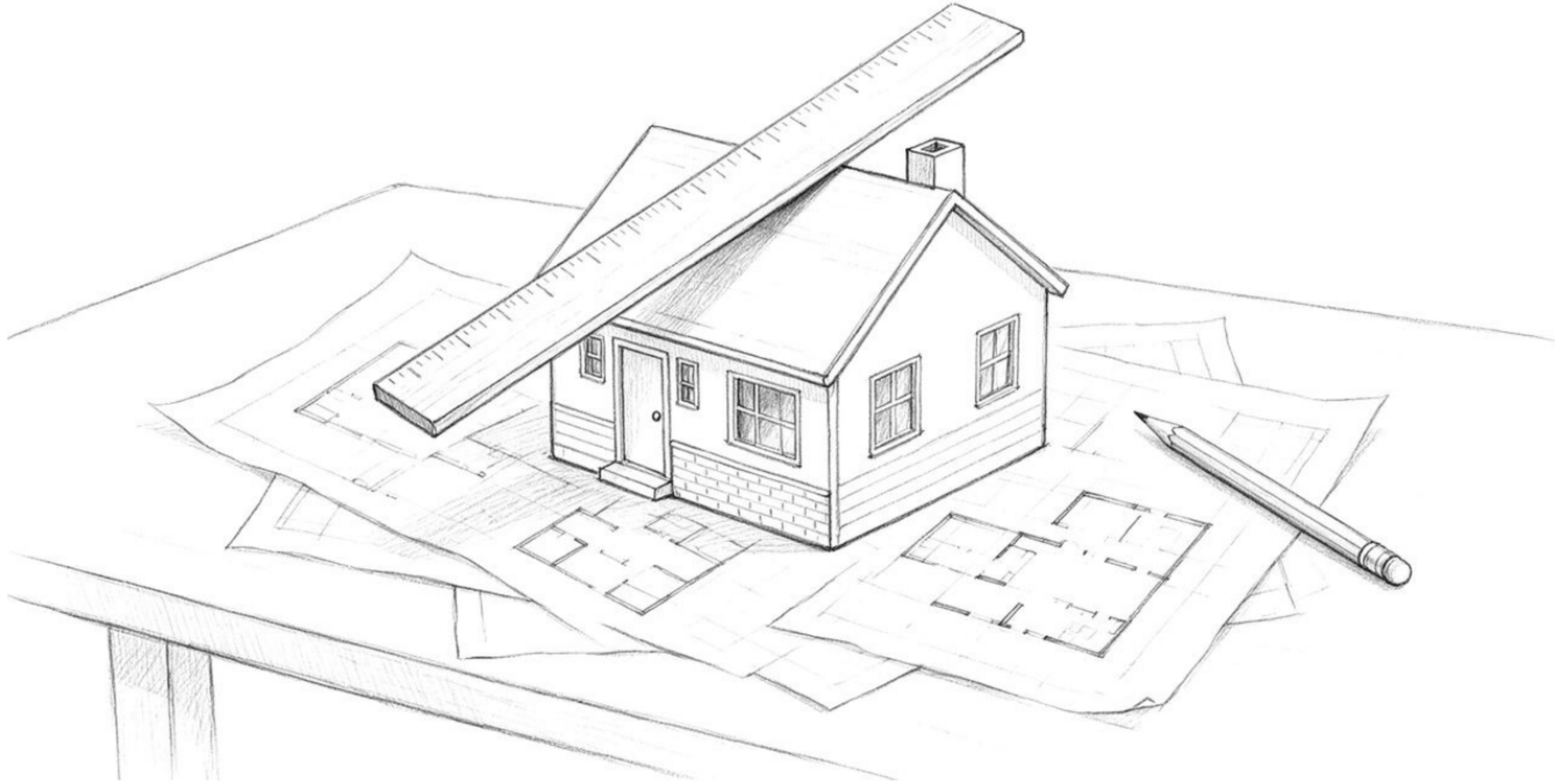


No labeled data — just consequences, tried and adjusted

Reinforcement learning: reward and penalty shape behavior



This is RLHF — the step that made ChatGPT helpful



Features and the train/test split

Features are the columns, each row is one example

sq ft	bedrooms	location

label:
sale price

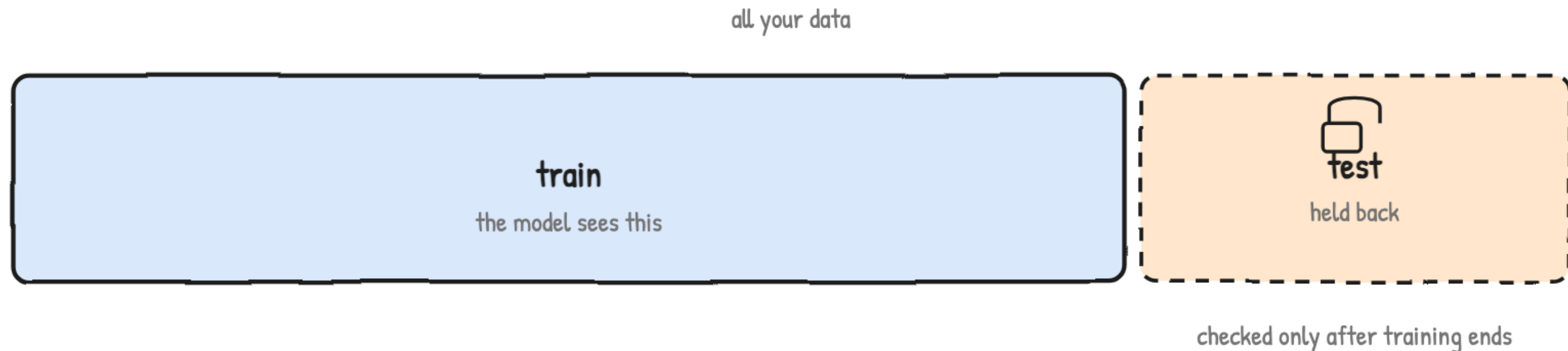
The measurable inputs a model learns from

Studying for the exam using the actual questions

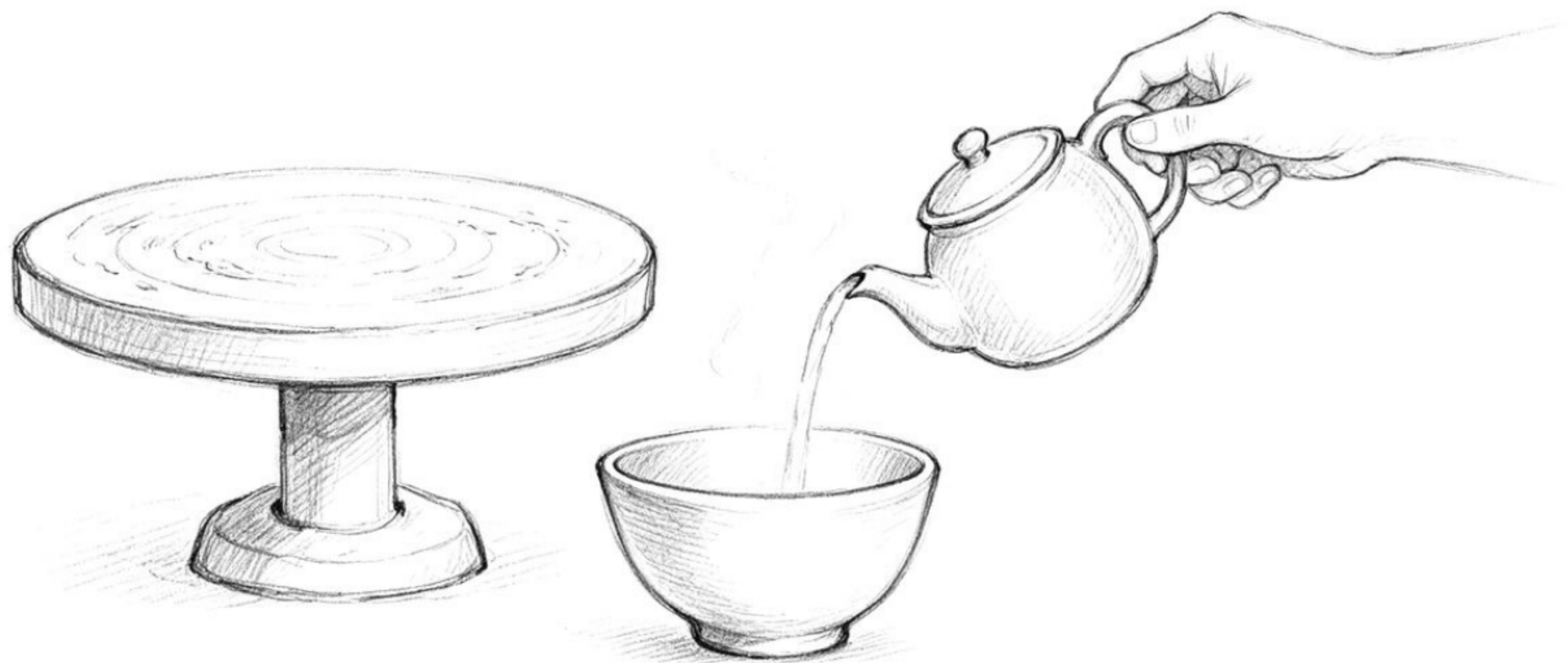


You'll score high and prove nothing

Hold back data the model never sees during training



The gap between the two scores tells the truth



Training vs inference – why the model doesn't learn
from your chats

Training adjusts the model; inference just uses it

TRAINING

sees examples
compares guess to the answer
updates the numbers

once, offline —
hours or days

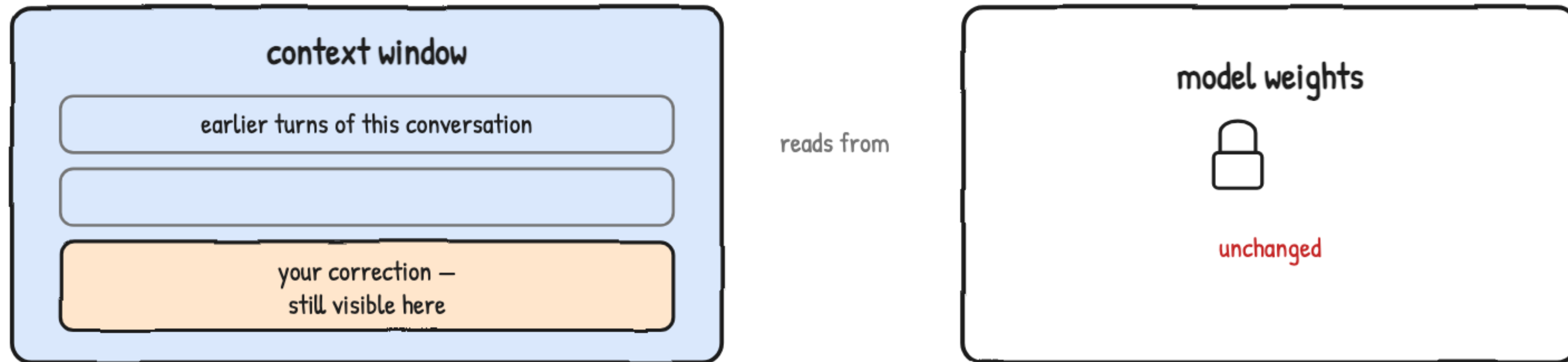
INFERENCE

input →
fixed numbers →
output

every single time
you send a message

One happens once — the other happens every time you type

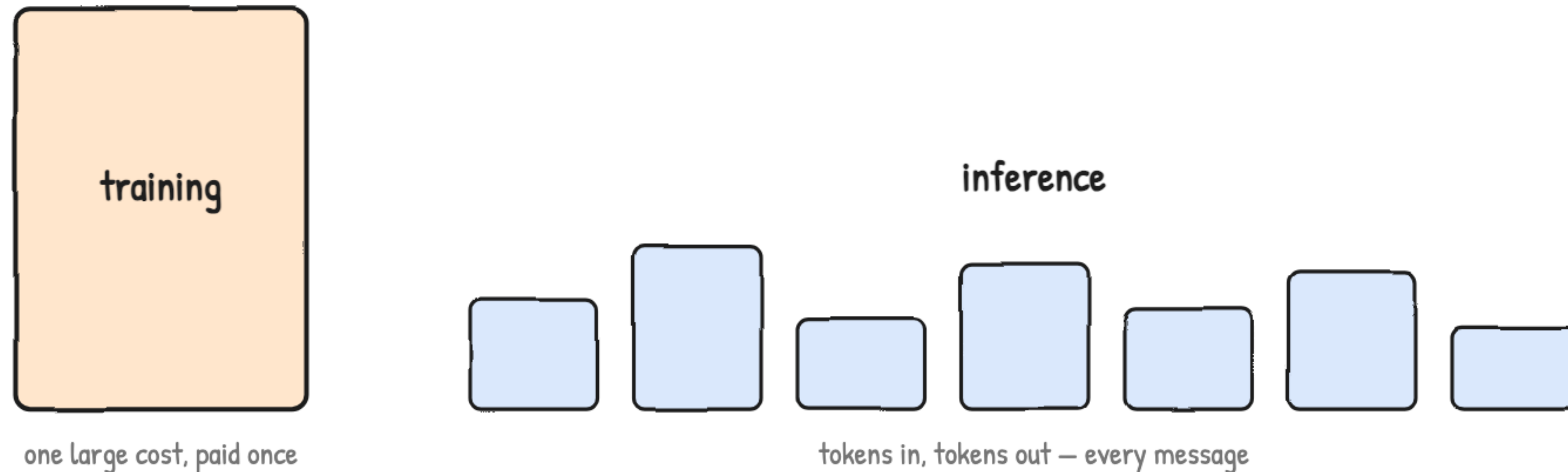
It looks like it's learning — it's only reading the page



conversation ends → window empties → model exactly as before

The correction sits in the window, never in the numbers

Training is a one-time bill; inference is metered, per call



That's what you're actually paying for, in tokens



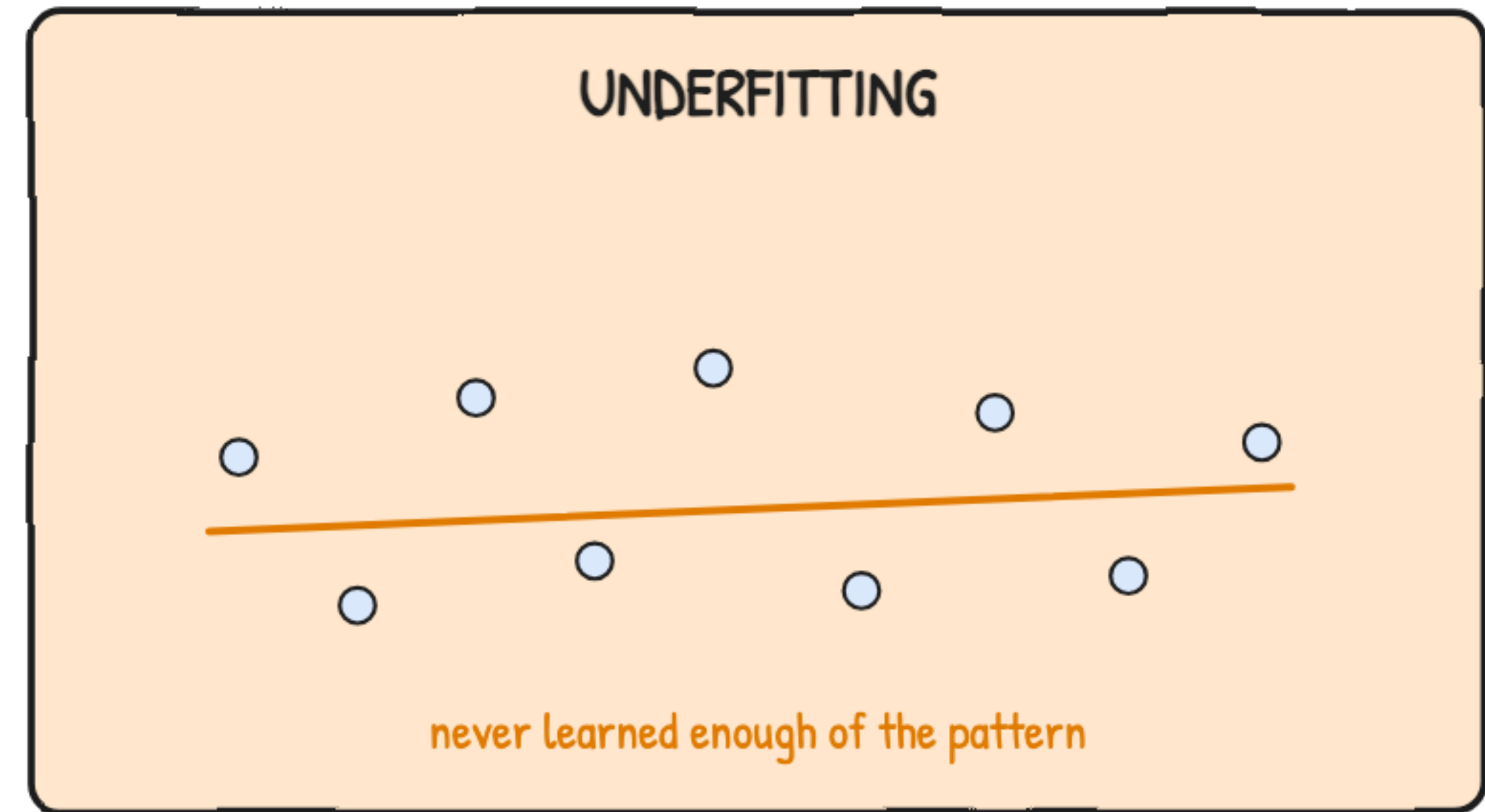
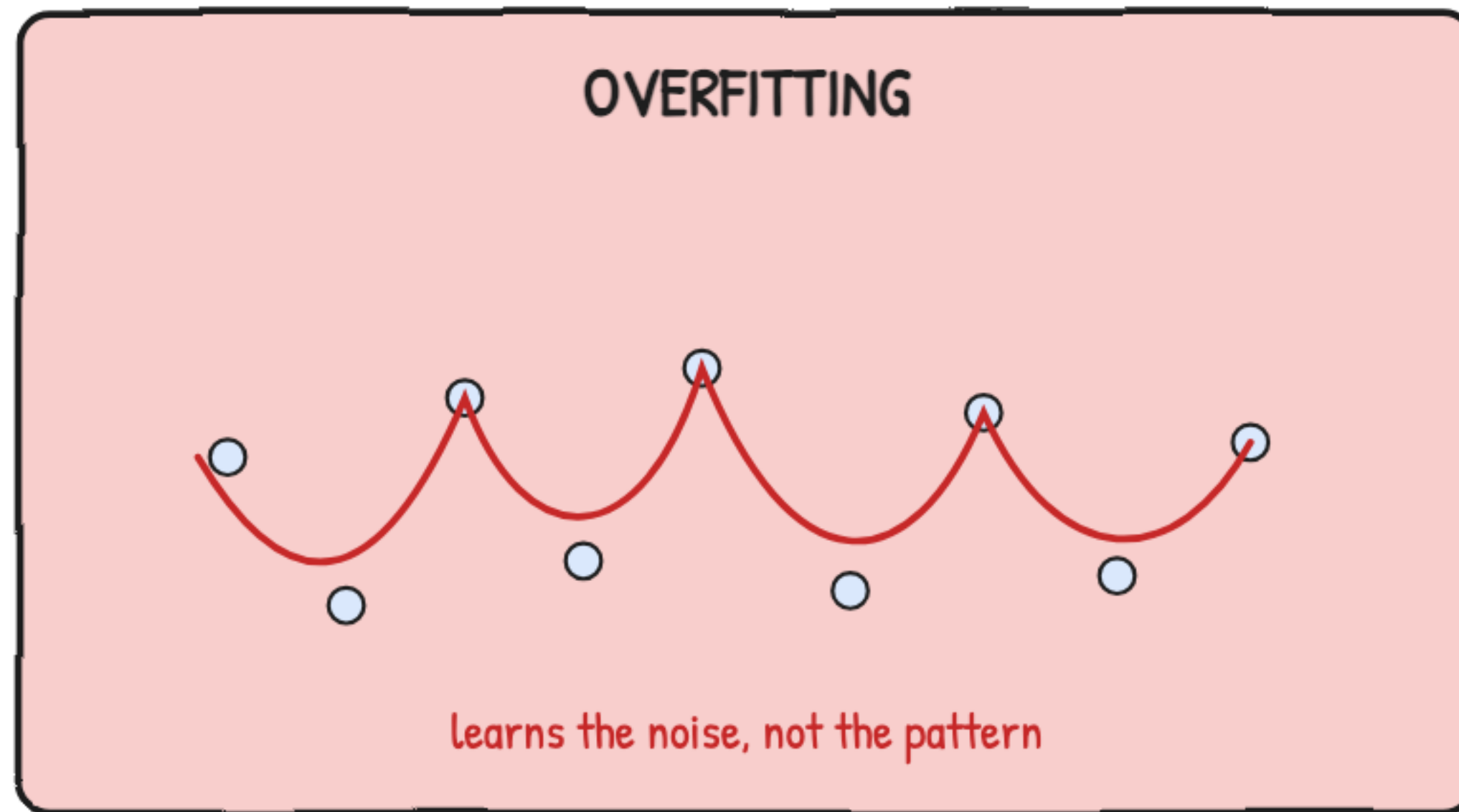
**When learning goes wrong: overfitting, underfitting,
and bias-variance**

Memorized last year's exact exam paper, word for word



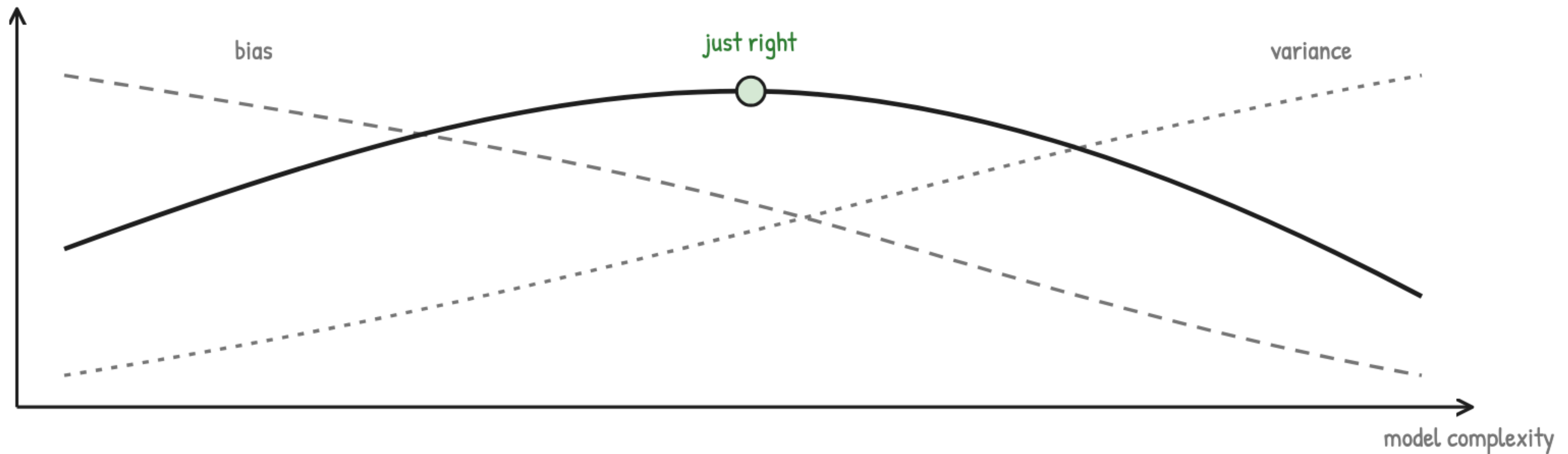
Aces a repeat, fails anything asked differently

One memorizes noise, the other never learns enough



Neither one learned the real pattern

The bias-variance trade-off: too simple or too sensitive



High bias underfits; high variance overfits

Hallucination is overfitting and underfitting, at internet scale

asked something outside
what it learned well

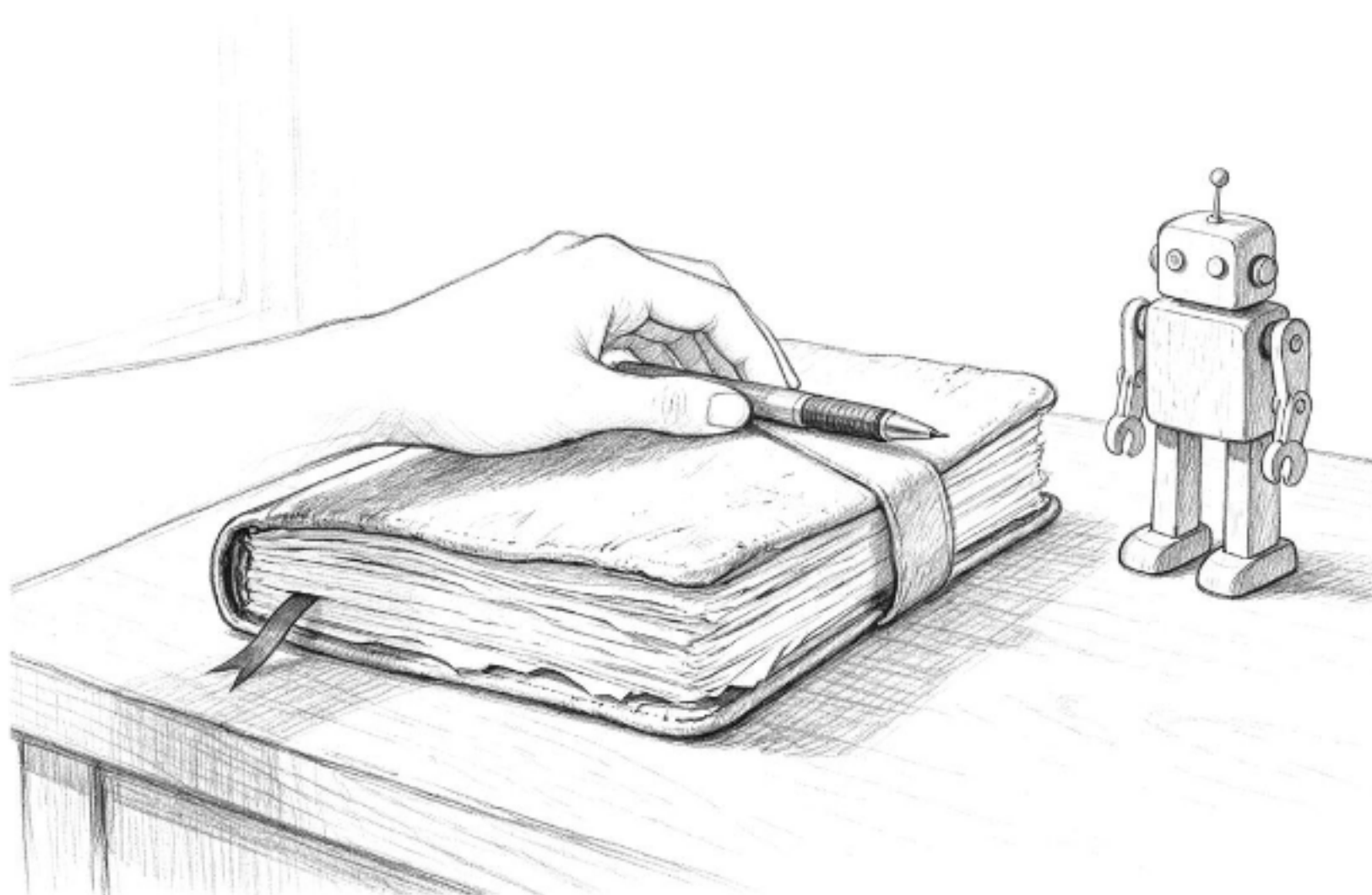
a fluent, confident-sounding
sentence — generated anyway

sounds right — isn't grounded in a real pattern

No real pattern to fall back on, so it guesses fluently

MODULE 3 TAKEAWAYS

You now know what training actually is — and isn't



- 1 three ways a machine learns — and where each one lives today
- 2 features & the train/test split — the one honest test
- 3 training vs inference — why your chats change nothing
- 4 overfitting, underfitting & bias-variance — the root of hallucination

Now open Module 3 · Lab. · Gourav Shah · School of DevOps & AI